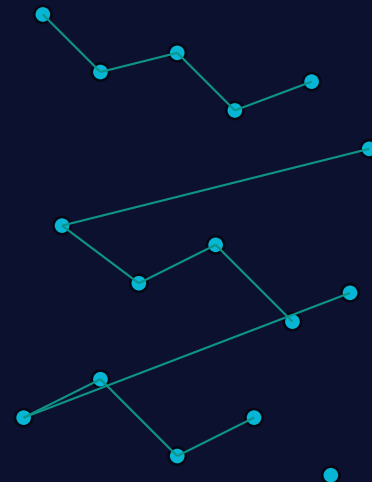


WHALE SHARK MOVEMENT & SST MODELING

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id: 108113_0

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Two Pillars of Progress

01

Data Processing

- ▶ Truncation by 7-day gaps
- ▶ Min 20 obs per block
- ▶ SSM + MPM fitting & diagnostics

02

Statistical Modeling

- ▶ LMM → Logit-GLMM → Beta GAMM
- ▶ SST as key predictor
- ▶ Movement persistence (γ) response

Truncation & Filtering Pipeline



Raw Tracking Data

Raw satellite tag positions

Irregular temporal spacing

Multiple individuals



Gap Filter (> 7 days)

Consecutive observations

Checked for gaps > 7 days

Blocks split at each gap



Min Block Size

Each block must have ≥ 20 observations

Rest are removed

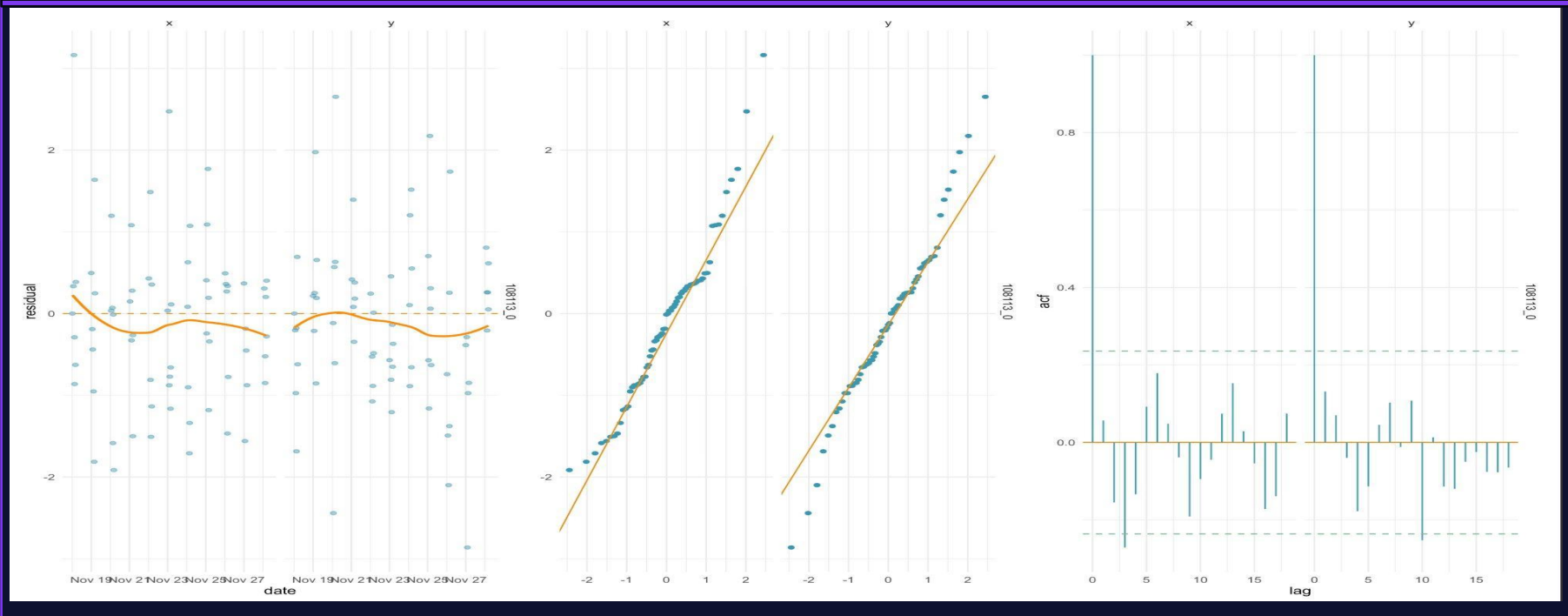
State-Space Model (SSM)

Regularizes irregular tracks $\rightarrow$ continuous position estimates

Movement Persistence Model (MPM)

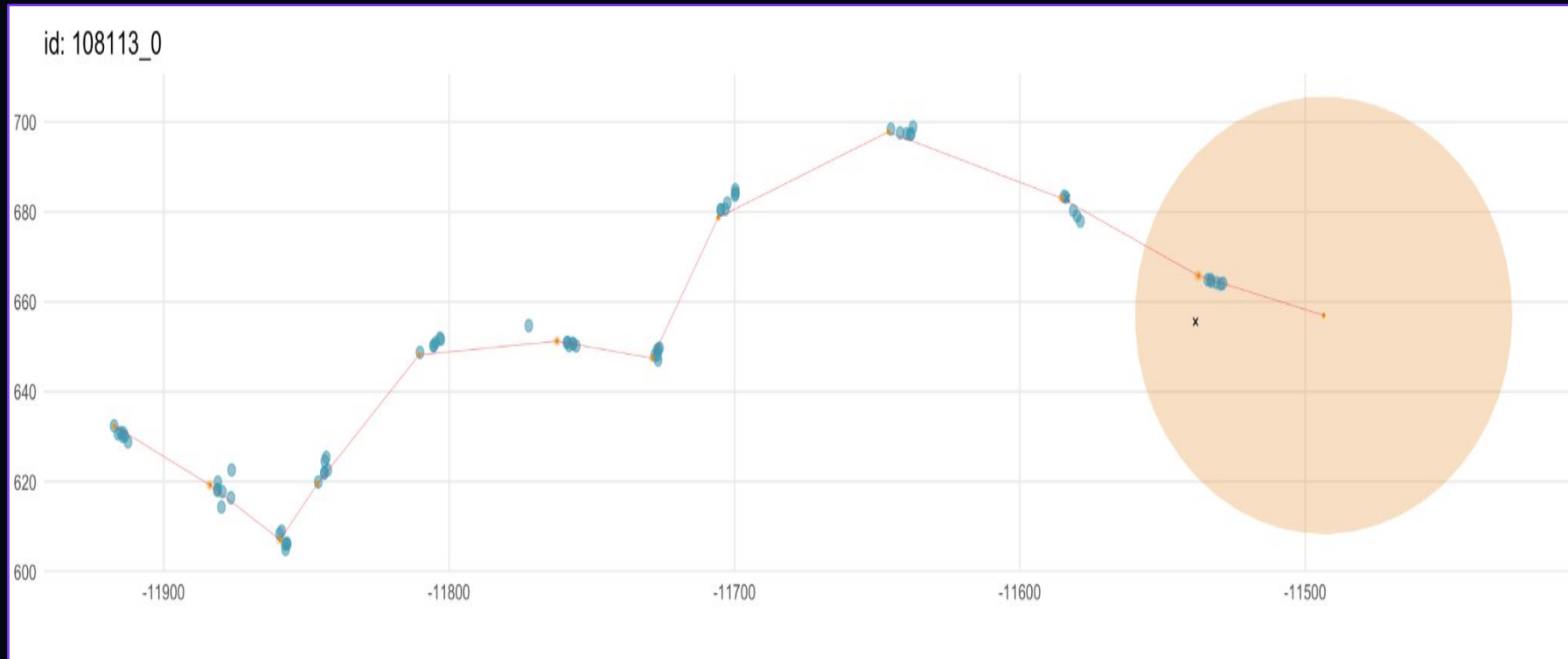
Extracts $\gamma \in (0,1)$ — directional persistence of movement per observation

SSM Diagnostic Plots — id: 108113_0

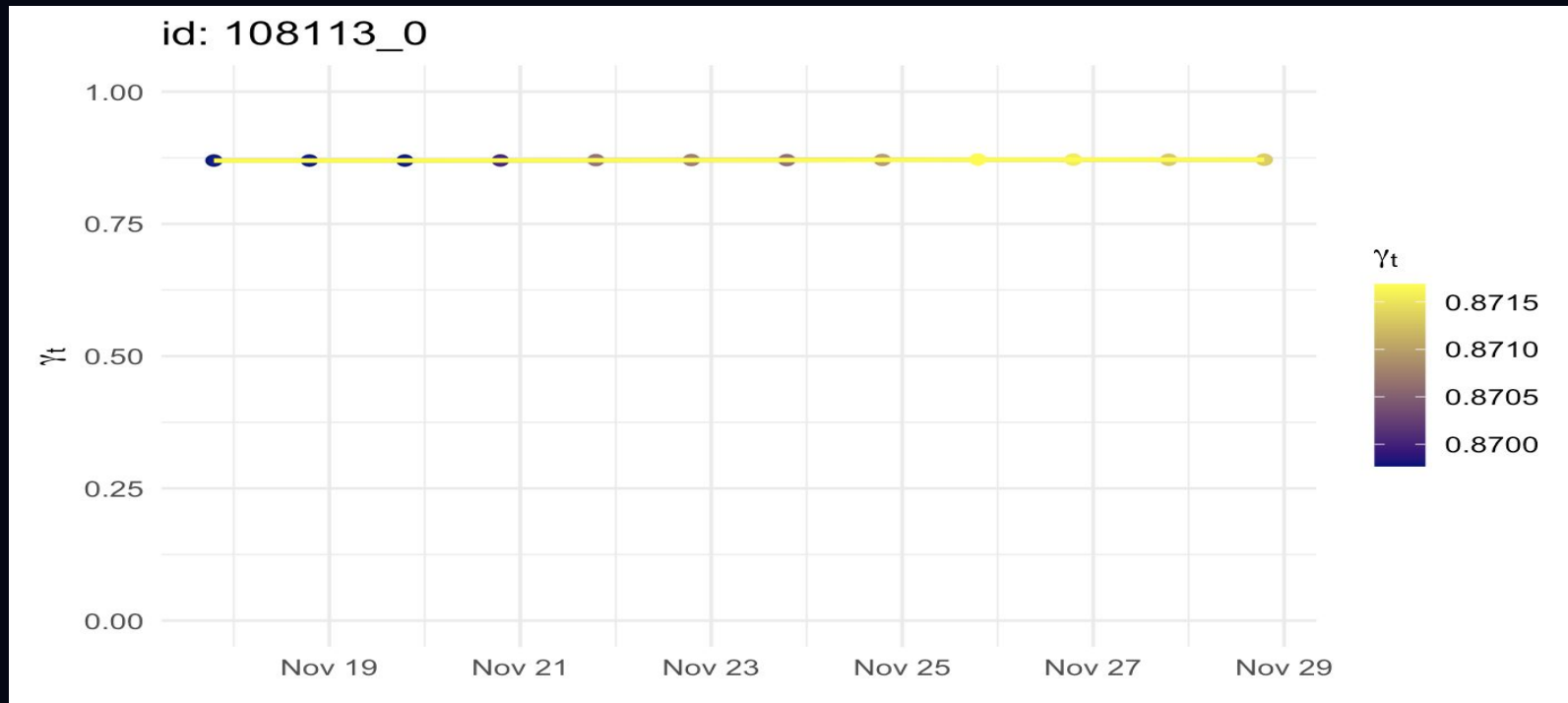


Residual vs. date · Q-Q plot · ACF — plotted for x & y coordinates separately

Observed vs. Predicted Trajectories



MPM Gamma Estimates for example track



Sea Surface Temperature as Environmental Predictor

Extraction Method

Matched SST to each **MPM observation** point

Temporal alignment: **daily** satellite-derived SST

For each truncated shark segment, used (latitude, longitude, date) as the lookup key

Extracted the corresponding SST values using the **rerddapXtracto** package

Why SST?

- Key driver of marine animal movement behaviour
- Whale sharks thermally sensitive — known to follow warm fronts
- Hypothesized: high SST → area-restricted search → low γ
- Readily available, global coverage

Linear Mixed Model (LMM, linear + random effect)

Formula: gamma ~ sst + (1 | id)

Data: analysis_df

REML criterion at convergence: -1425.6

Scaled residuals:

Min	1Q	Median	3Q	Max
-4.6193	-0.1435	0.0248	0.3720	3.7383

Random effects:

Groups	Name	Variance	Std.Dev.
id	(Intercept)	0.04323	0.2079
	Residual	0.01970	0.1404

Number of obs: 1511, groups: id, 53

Fixed effects:

	Estimate	Std. Error	t value
(Intercept)	1.192986	0.088106	13.540
sst	-0.018838	0.003285	-5.735

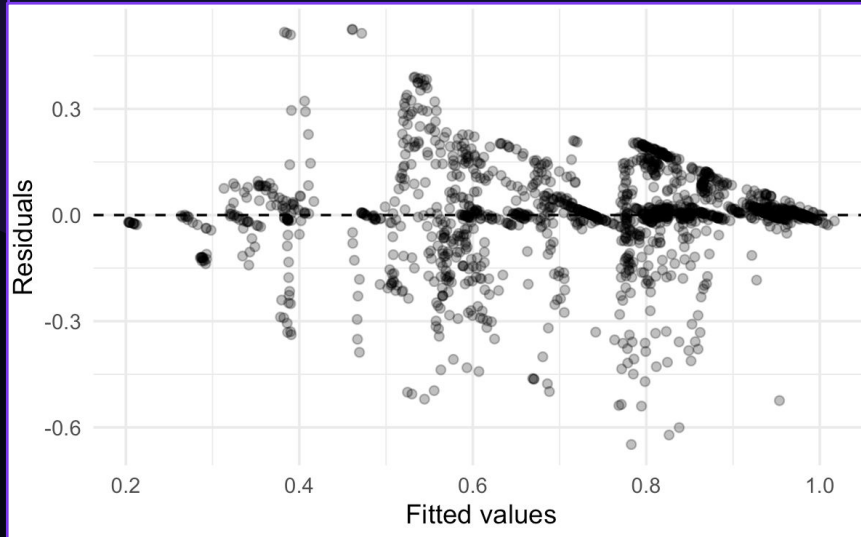
Correlation of Fixed Effects:

(Intr)

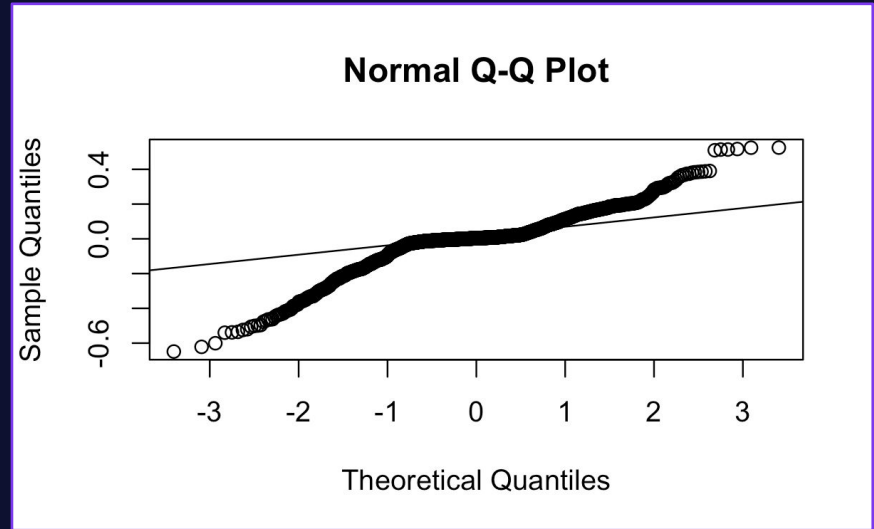
sst -0.945

- Higher SST is associated with lower movement persistence (γ).
- Substantial variability across sharks (random intercept effect).
- Gaussian LMM used as a baseline; assumptions not fully satisfied.

LMM Diagnostics



● Residual vs Fitted - heteroscedasticity



● Normal Q-Q Plot -notable departures in both tails

Logit Transformed LMM

```
Formula: gamma_logit ~ sst + (1 | id)
Data: model_df
```

REML criterion at convergence: 4453.7

Scaled residuals:

	Min	1Q	Median	3Q	Max
	-4.7861	-0.2027	0.0065	0.2655	3.9937

Random effects:

Groups	Name	Variance	Std.Dev.
id	(Intercept)	2.3243	1.5246
Residual		0.9667	0.9832

Number of obs: 1511, groups: id, 53

Fixed effects:

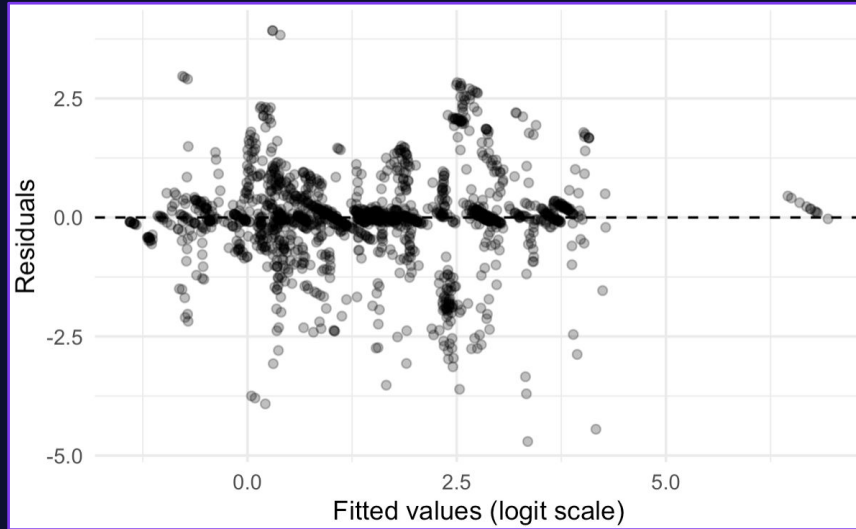
	Estimate	Std. Error	t value
(Intercept)	5.50112	0.62107	8.857
sst	-0.16000	0.02304	-6.944

Correlation of Fixed Effects:

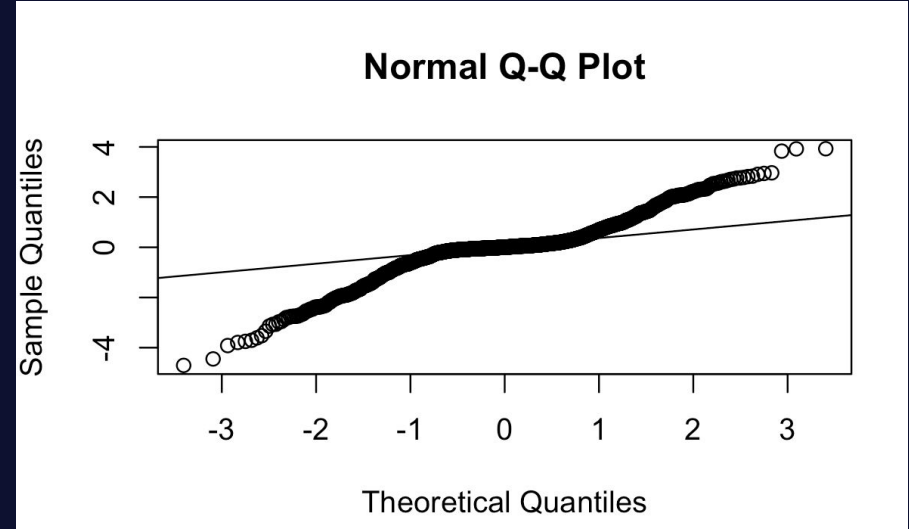
	(Intr)
sst	-0.940

- Logit transformation accounts for y being bounded in $(0,1)$.
- Negative association between SST and movement persistence remains ($\beta \approx -0.16$).
- Substantial between-shark variability (random intercept effect).

Logit Transformed LMM



● Residual vs Fitted - heteroscedasticity



● Normal Q-Q Plot -notable departures in both tails

Beta GAMM (Beta Regression + Logit Link + Random Effect)

Family: Beta regression(7.972)

Link function: logit

Formula:

`gamma_beta ~ s(sst, k = 10) + s(id, bs = "re")`

Parametric coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.0757	0.1501	7.165	7.76e-13 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:

	edf	Ref.df	Chi.sq	p-value
s(sst)	5.01	6.196	42.94	<2e-16 ***
s(id)	50.49	52.000	2124.36	<2e-16 ***

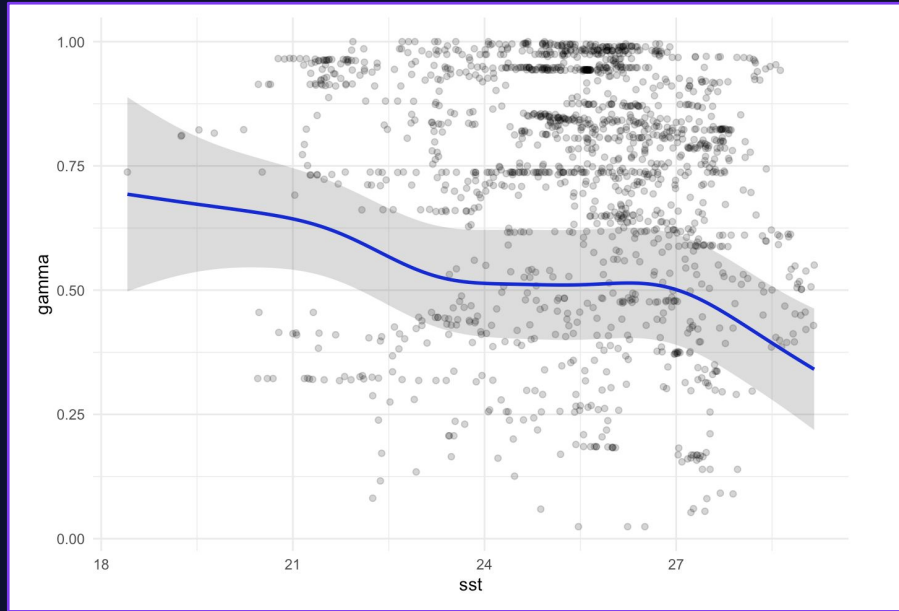
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) = 0.615 Deviance explained = 69.9%

-REML = -1136.9 Scale est. = 1 n = 1511

- Models y directly on (0,1) using a Beta distribution with a logit link.
- Nonlinear SST effect captured with a smooth term.
- Strong individual variability modeled with a random effect for shark ID.

Beta GAMM (Beta Regression + Logit Link + Random Effect)



- The SST- γ relationship is nonlinear.
- Movement persistence generally decreases as SST increases.
- Uncertainty increases at the SST extremes.

Model Selection Journey

M1

✗ Assumptions Violated

Gaussian LMM

Random intercept for shark ID

Issues: heteroscedasticity +
non-normal tails

M2

⚠ Partial Improvement

Logit-transformed LMM

$\text{Logit}(y) \sim \text{SST} + (1|\text{track})$
SST coeff = -0.16 (significant)

Still: heteroscedasticity +
tail departures

M3

✓ Best Fit

Beta GAMM

$\text{gamma} \sim s(\text{SST}) + (1|\text{shark})$
Models y directly on $(0,1)$

Allows nonlinear SST effects
via splines

Key Findings

-0.16

SST coefficient

(logit-LMM scale)

2.32

Between-track random

intercept variance

0.97

Residual variance

(logit-LMM)

Interpretation — Beta GAMM

- Higher SST is associated with lower movement persistence (γ) — consistent across all model frameworks.
- Large between-track variance suggests substantial individual variation in baseline movement behaviour.
- Beta GAMM with smooth SST term best captures the nonlinear relationship and bounded $[0,1]$ nature of γ .

Where We Are & Where We're Going

✓ Key Findings

Higher SST → lower movement persistence (γ)

Consistent across LMM, Logit-LMM, Beta GAMM

Large between-individual variation in baseline γ

Beta GAMM best matches data structure (bounded response, nonlinear covariate)

→ Next Steps

01 Refine Beta GAMM — additional smooth terms (depth, seasonality?)

02 Explore spatial autocorrelation in residuals

03 Consider additional environmental covariates (Chlorophyll)